Problem FoL.1 ... jumping dog

An escaping prisoner needed to jump from one rooftop to another one because it's just th
hing escaping prisoners do. The first building, from which he jumped, is $H = 16\,\mathrm{m}$ *tall an*
he second one is $h = 11.6\,\mathrm{m}$*, the buildings are* $d = 4\,\mathrm{m}$ *apart. The velocity of the prisoner a*
he moment of jump is $v = 3.8\,\mathrm{m{\cdot}s^{-1}}$ *and he is jumping parallel to the sufrace. Determine th*
missing/redundant distance (using the -/+ signs) after the jump relative to the rooftop of th
econd building. Air resistance is negligible.Assume gravitational acceleration $g = 9.81\,\mathrm{m{\cdot}s^{-2}}$

Kiki has found and remade this HRW problem while drinking tea

Problem FoL.2 ... source of radiation

Compute the wavelength of radiation passing through an optical grid. The distance of th optical grid from the projection screen is $2.0\,\mathrm{m}$*, the distance between* 0th *and* 2nd *maximum is* $6.0\,\mathrm{cm}$*. Period of the optical grid is* $5.0 \cdot 10^{-6}\,\mathrm{m}$*. The result should be in nanometers.*

Monika couldn't believe her eyes

Problem FoL.3 ... fluctuational

Quantum electrodynamics which connects quantum theory with the relativity brought revolu
ion into our understanding of the vacuum. Vacuum isn't emptiness and nothingness anymore
t's a state with the lowest energy. The lowest possible energy state cannot be zero since w
ave to take into account the uncertainty principle – which leads us to the fluctuations of th
acuum. The strongest mechanical demonstration of quantum fluctuations is the attraction for
e between two mirrors (two parallel metallic plates without any charge) separated by a narrow
ap. Around the system, there are waves with all frequencies but inside the gap, there are onl
uch waves which correspond to the resonance frequency of the gap. This results in a small
et measurable force which pushes the mirrors closer to each other. It's magnitude depends o
he area of the mirrors S (it's natural to assume linear dependence here) and the width of th
ap d. There are two fundamental constants in the relation, c and h:

$$F = K c^{\alpha} h^{\beta} d^{\gamma} S\,.$$

Using dimensional analysis, determine the coefficients α, β and γ and the force for the value
$S = 1\,\mathrm{cm}^2$ and $d = 1\,\mu\mathrm{m}$. K is a dimensionless constant, its value is $K = \pi/480$ as can b
derived with precise calculations.

Zdeněk went through his old excercise book and was surprised to no end

Problem FoL.4 ... czech energetics

Imagine you transform a 500 *Euro banknote into energy. How many times more valuable does i become? The banknote weights* 1.1 g, 1 kWh *of electric energy costs* 20 *cents. Round the result to tens.* *Try to discover the real value of money*

Problem FoL.5 ... fall one

What is the stabilized velocity of a falling leaf, if the leaf has enough time to reach an equ
ibrium. Consider the grammage of the leaf similar to be to that of office paper ($80\,\mathrm{g{\cdot}m^{-2}}$
nd that it has the shape is of a hemisphere. Consider Newton's relation for air resistance an
ake air density to be $\varrho = 1.29\,\mathrm{kg{\cdot}m^{-3}}$ *and the drag coefficient* $C = 0.33$*. Assume gratitationa*
cceleration $g = 9.81\,\mathrm{m{\cdot}s^{-2}}$*.* *Those who are afraid of falling leaves have a guilty conscience*

Problem FoL.6 ... choo choo train

A little train named Denis decided to go through one turn repetitively, speeding up till h derails. The radius of the turn is $R = 190\,\mathrm{m}$, tilt of the rail is $\alpha = 5°$, the rail width is $d = 1.4\,\mathrm{m}$ and the center of the mass of the train is $h = 1.6\,\mathrm{m}$ from the rails. What's the difference betwee minimal and maximal speed with which the train could go through the turn without derailing The result should be in kilometres per hour. The gravitational acceleration is $g = 9.81\,\mathrm{m{\cdot}s^{-2}}$ The train moves horizontally.

Jakub wants a driving licence for a train

Problem FoL.7 ... collision!!!

A positron and a helium nucleus are approaching each other, both are moving in a straight line with the same speed $v = 2\,000\,\mathrm{km{\cdot}s^{-1}}$. *What will the distance be between them in the moment when the positron stops (in the reference frame of the lab)? Express in* pm. *Solve it clasically.*

Tomáš Bárta wanted to assign a particle physics task

Problem FoL.8 ... semicircular analyzer

There is a specific instrument to analyze the distribution of the electrons flying out of th experimental device. It consists of two semicircles of $R = 20\,\mathrm{cm}$ radius, with a magnetic fiel between them. A half of the cut edge of the semicircle is the detector, while in the second hal there is, in the distance $d = 15\,\mathrm{cm}$ from the center, a crevice through which the electrons ar jected into the space with magnetic field. Assume the electrons are moving at nonrelativisti speed. Derive the ratio η of maximal and minimal velocity of the electrons, measurable wit this device. *Aleš wrote down the first thing which came onto his mind*

Problem FoL.9 ... upgrade

Lately, CERN upgraded their super collider. They increased the energy of the accelerate protons from $3.5\,\mathrm{TeV}$ *to* $7\,\mathrm{TeV}$*. We would like to know how it affected the velocity of th protons. Invariant mass of the proton is* $938\,\mathrm{MeV}/c^2$*.*

Jakub was a bit interested by this question

Problem FoL.10 ... let there be equality

In the container, there is a mixture of $m_1 = 50\,\mathrm{g}$ of iodide ^{131}I and $m_2 = 20\,\mathrm{g}$ of strontium ^{90}Sr. How long will it take to have the same number of atoms of each element in that container?

Taken from Chemistry Olympics

Problem FoL.11 ... apple cider

Consider a homogeneous cylinder with both pedestals of mass $M = 100\,\mathrm{g}$ *and height* $H =$
$= 20\,\mathrm{cm}$. *The cylinder is hanging in the air so its axis of symmetry is identical with th*
direction of the gravitational acceleration. At first, the cylinder is full of apple cider of mas
$m_0 = 500\,\mathrm{g}$. *Let's make a small hole in the bottom of the cylinder, so the cider runs out of it*
Compute the height of the surface of the cider for the lowest position of the center of mass o
he system. Give the result in cm. *Domča likes fall fruits and their derivatives*

Problem FoL.12 ... Wien's filter

If you ever need to filter particles of a certain velocity from others, you can simply do tha using perpendicular magnetic and electric fields. Let's assume both of them are homogeneou and oriented so that the particle flying through them flies in a perpendicular direction to bot of them. The forces with which the fields are acting on the particles are anti parallel. Wha is the velocity the particle must have in order to reach the detector or in other words to pas the velocity filter which is aligned with the original direction of motion of the particle? Electri field intensity is $E = 9 \cdot 10^3\,\mathrm{V{\cdot}m^{-1}}$ *and the magnetic induction is* $B = 3 \cdot 10^{-2}\,\mathrm{T}$.

From the experimental physicist's life

Problem FoL.13 ... pseudo-ice tea

Kiki felt like having some ice tea, but she didn't know how to prepare it. So she tried to d
t in the following way: she put $m_w = 250\,\mathrm{g}$ of water at temperature $t_w = 20\,^\circ\mathrm{C}$ into a kettl
long with $m_i = 350\,\mathrm{g}$ of ice at temperature $t_i = 0\,^\circ\mathrm{C}$. After some time (precisely when th
ontents of the kettle reached the thermal equilibrium), she figured out that she might want t
urn on the kettle. How long will it take to boil the water in the kettle, whose power input an
fficiency are known to be $P = 1.8\,\mathrm{kW}$ and $80\,\%$ respectively?

Kiki was drinking tea while thinking of problems for the competition

Problem FoL.14 ... climbing acetone

Find the elevation of acetone at temperature $20\,^{\circ}\mathrm{C}$ *in a capillary with diameter* $0.6\,\mathrm{mm}$. *Surfac*
ension of acetone is $0.023\,4\,\mathrm{N{\cdot}m^{-1}}$. *Your result should be stated in centimetres.*

Kiki remembered a set of problems from physical chemistry

Problem FoL.15 ... the round one

magine a small glass ball with the property that it focuses parallel rays of light inciden
erpendicularly on its surface onto its back side. What is the refractive index of the glass th
all is made of? Work in paraxial approximation. The result should be expressed as a multipl
f the refractive index of ball's surroundings.

Zdeněk found a very suspicious ball in his drawer

Problem FoL.16 ... the bulb and the capacitor

When Oliver was in New York, he bought a GE 31546 60A1 P VRS ES 110 120V BE typ
ight bulb. When he came back to Czech republic, he didn't really want to throw it away, s
n order for the bulb to work in Czech republic he had to connect it in series with an idea
apacitor. What should be the capacitance of this capacitor so that the correct nominal value o
he voltage across the light bulb is obtained? The alternating current in European distributio
etwork has frequency 50 Hz *and the household outlet voltage in Czech Republic is* 230 V. *D*
ot take into account the internal resistance of the source. The answer should be stated in μ*F*

Jimmy was inspired by a friend of his, who bought a wrong bulb

Problem FoL.17 ... the rod on the wires

Consider a rod of mass $m = 97\,\mathrm{kg}$ *suspended on two steel wires* Q *and* W *with equal radi* $= 1.3\,\mathrm{mm}$ *and elastic moduli* $E = 210 \cdot 10^9\,\mathrm{Pa}$ *in such a way that the rod is parallel to th horizontal, as we can see in the figure. Wire* Q *was originally (before we used it to suspen the rod)* $l_0 = 2.7\,\mathrm{m}$ *long while wire* W *was by* $\Delta l = 2\,\mathrm{mm}$ *longer. Let us denote the horizonta distances of the wires from rod's center of mass by* d_Q, d_W*, as we did in the figure. What is th ratio* d_Q/d_W*? The acceleration due to gravity is* $g = 10\,\mathrm{m{\cdot}s^{-2}}$*. The result should be stated u to two significant figures. Assume that the radii of the wires stay unchanged.*

Domča was playing with ropes made of construction stee

Problem FoL.18 ... totem

In one of the Plitvička lakes, there is a place where the lake is 2 m *deep and a vertical totem sticks out of the water reaching to the height of one meter above the water level. The rays coming from the setting Sun located at the altitude of* 30° *above the horizon are incident on the totem. How long is the shadow (in meters) cast by the totem onto the bottom of the lake?*

Dominika was watching Karl May's classics

Problem FoL.19 ... solar mania

The Sun shines onto the Discworld at an angle of 39° *measured relative to its horizon. In such a situation the illuminance cast on its surface is* $E_1 = 80 \cdot 10^3\,\mathrm{lx}$. *Find the illuminance cast on the surface of the Discworld when the Sun is only* 30° *above the horizon.*

f(Aleš) was lacking light

Problem FoL.20 ... a decaying one

Imagine we have a 20.0 g *sample of an unknown radioactive element whose nuclei are know*
to contain 232 *nucleons. In the sample,* $2.12 \cdot 10^{11}$ *decay events per minute are observed t*
occur. Find the half-life of the element. The product of the decay is assumed to be stable

f(Aleš) having fun during a lecture on nuclear physics

Problem FoL.21 ... thermo-sphere

t is widely known that ordinary light bulbs emit a substantially larger portion of their radiatio
n infrared than they do in visible. Imagine our heating went on strike and we wish, using ou
ight bulb, to warm up our hands from the temperature $T_1 = 15\,^{\circ}\mathrm{C}$ *to* $T_2 = 35\,^{\circ}\mathrm{C}$. *We go o*
nd cover the bulb entirely by our hands, thus exploiting all the thermal power emitted by th
ncandescent tungsten filament. Find the time needed to bring our frozen hands to the require
emperature given the temperature of the tungsten filament is known to be $T_W = 3\,000\,\mathrm{K}$, *it*

ength $l = 10^{-1}\,\mathrm{m}$ *and its diameter* $d = 10^{-4}\,\mathrm{m}$. *We estimate the mass and the heat capacit* *f our hands as* $m = 1\,\mathrm{kg}$ *and* $c = 3\,000\,\mathrm{J{\cdot}K^{-1}{\cdot}kg^{-1}}$ *respectively.*

Mirek and his striking heating

Problem FoL.22 ... analogue hydrometer

Due to its higher density, cold water stays close to the bottom of a rectangular vessel which i
illed up to the height of $h = 30\,\mathrm{cm}$. *We assume that the density of water in the vessel grow*
inearly with increasing depth – at the water level, the density is equal to $\varrho_l = 996\,\mathrm{kg{\cdot}m^{-3}}$
while the density ϱ_b *at the bottom of the vessel is unknown. Determine this density using th*
act, that a homogeneous rod with density $\varrho_r = 997\,\mathrm{kg{\cdot}m^{-3}}$ *and length* h *immersed in the wate*
nd fixed by one of its ends at the water level makes an angle of $\varphi = 60^\circ$ *with the vertica*

Mirek was thinking about alternative measuring instruments

Problem FoL.23 ... pressurized box

Assume we have $n = 1\,\mathrm{mol}$ *of carbon dioxide* (CO_2) *in a closed vessel with volume* $V = 1\,\mathrm{l}$. *Th vessel is in thermal equilibrium with its surroundings at temperature* $T = 297\,\mathrm{K}$. *How does th estimate of pressure in the vessel based on the ideal gas law (denote* p_{id}) *differ from the th estimate based on the van der Waals equation* (1) *of state for non-ideal fluid (denote* p_{Waals})?

$$\left(p_{Waals} + \frac{n^2 a}{V^2}\right)(V - nb) = nRT \qquad (1$$

Determine $(p_{id} - p_{Waals})/p_{id}$. *Use the following values of* a, b *for* CO_2:

$$a = 0.3653\,\mathrm{Pa{\cdot}m^6{\cdot}mol^{-2}}\,,$$
$$b = 4.280 \cdot 10^{-5}\,\mathrm{m^3{\cdot}mol^{-1}}\,.$$

The molar gas constant is $R = 8.31\,\mathrm{J{\cdot}K^{-1}{\cdot}mol^{-1}}$.

Karel wanted to mention van der Waals gas

Problem FoL.24 ... world champion in high jump

Brian Griffin (a dog) has one spring mounted to each of his hind legs. Each of these springs ha
an unstretched length of $l = 0.5\,\mathrm{m}$ *and spring constant* $k = 3 \cdot 10^5\,\mathrm{kg{\cdot}s^{-2}}$*. The dog then jump*
to a vertical height of $h = 10\,\mathrm{m}$ *and when he falls down, the springs become hooked into th*
ground so the dog starts oscillating. What is the amplitude of the undamped oscillations for
dog weighing $m = 500\,\mathrm{kg}$*? Assume gravitational accleration* $g = 9.81\,\mathrm{m{\cdot}s^{-2}}$*.*

Mirek envied Brian his funny means of transport

Problem FoL.25 ... pulleys

Calculate the vertical acceleration of the weight with mass m_1 in the pulley system depicted i
he figure, assuming that the system is initially at rest. The masses of individual weights ar
$m_1 = 400\,\mathrm{g}$, $m_2 = 200\,\mathrm{g}$, $m_3 = 100\,\mathrm{g}$. Use $g = 10\,\mathrm{m{\cdot}s^{-2}}$ as the value for the acceleration due t
gravity. Pulleys and strings are weightless and friction can be neglected.

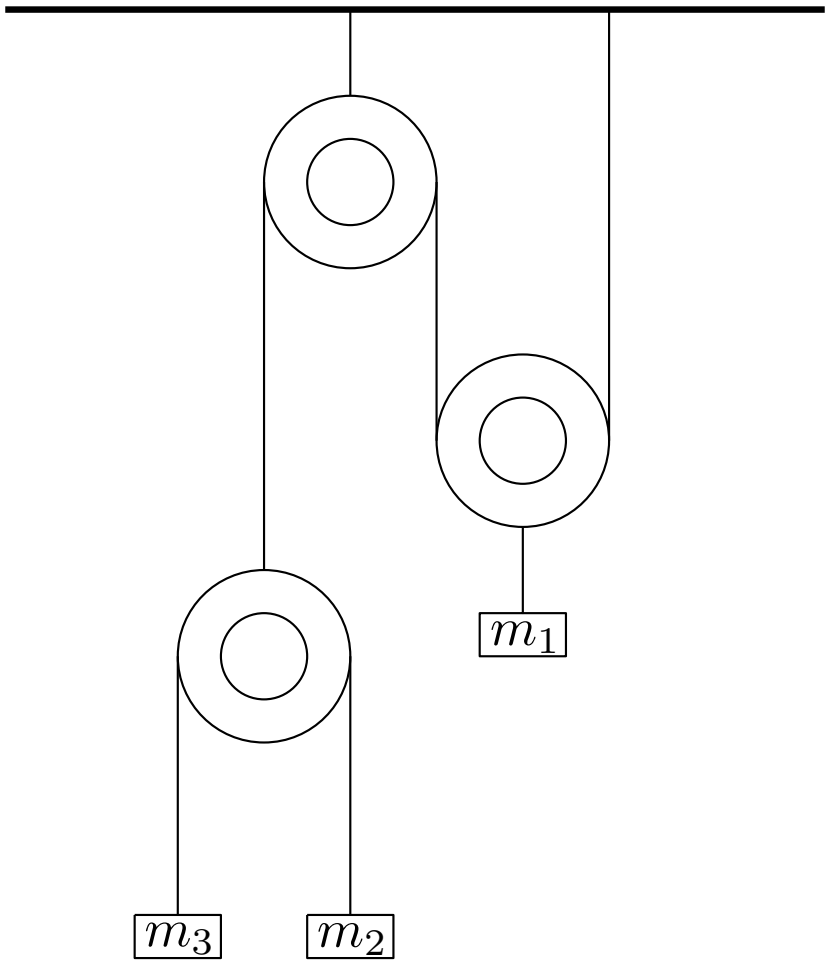


Fig. 4: Pulleys machine

Mirek was amazed at simple machines

Problem FoL.26 ... boxed light

How many photons of blue light does it take to achieve pressure $p = 1\,\mathrm{bar} = 10^5\,\mathrm{Pa}$ *in an empt*
ube-shaped box? The wavelength of the photons is $\lambda = 450\,\mathrm{nm}$ *and the edge length of th*
ube is $a = 10\,\mathrm{cm}$*. The interior of the cube is perfectly reflective.*

Thermodynamics class acted upon Kare

Problem FoL.27 ... enlightened carriage

A carriage of mass $m = 100\,\mathrm{g}$ can move along rails without friction. Assume there is a vertica mirror attached to one side of the carriage. We then focus all the light emitted by a light bul with power $P = 60\,\mathrm{W}$ into a ray which is incident perpendicularly onto the mirror. Assumin that the carriage was initially at rest, how long will it take the carriage to travel a distanc $= 1\,\mathrm{m}$ (in seconds)? You can assume that the mirror is perfectly reflective and that the whol power of the light bulb transferred into radiation.

Jakub wanted a contactless turbo-propulsion

Problem FoL.28 ... Earth-cylinder

Assume that the Earth turned into an infinite cylinder whose radius and density are the sam as those of our real Earth with the distance to the Moon also remaining unchanged. What wi be the speed of the Moon (which remains spherical) in its orbit around the cylinder? Assum that the Earth's radius and density are $6\,378\,\mathrm{km}$ *and* $5\,515\,\mathrm{kg{\cdot}m^{-3}}$ *respectively. Also conside the Moon's orbit around the Earth to be circular.*

Tomáš Bárta imagining alternative universes

Problem FoL.29 ... pass me the hammer

An astronaut dropped a tool bag during one of his spacewalks, giving it an impulse. The bag then started receding from the spaceship along a straight line until it reached the distance $= 180\,\mathrm{m}$ from the ship. At this distance, the speed of the bag relative to the spaceship was ero. How many days did it take the bag to get to that spot? The mass of the bag and the mass of the spaceship are known to be $m_1 = 50\,\mathrm{kg}$ and $m_2 = 500\,\mathrm{kg}$ respectively. You can neglect any effects due to gravity of other bodies. *Lukáš remembered the infamous ISS bag*

Problem FoL.30 ... a messy one

A chemist was planning to prepare 100 ml *of potassium permanganate solution with concen ration* $c_1 = 0.000\,5\,\mathrm{mol{\cdot}dm^{-3}}$, *so he placed the required amount of potassium permanganat into a measuring flask, poured in distilled water and mixed the solution carefully. But then a accident happened and he spilled a part of the solution. He was very lazy and since nobod else saw him, he just topped the solution up to the original volume with distilled water an pretended that nothing happened. Then he started to feel bad about the whole incident, so h took a sample of his solution into a* 1 cm *long cuvette and put it into a spectrophotometer. Fro The subsequent measurement he found that having passed through the sample, the intensit of monochromatic light of wavelength* 526 nm *dropped by* 90 % *compared to its original value What was the volume of the solution the chemist spilled assuming that his measurements wer precise? Molar absorption coefficient of potassium permanganate for the above given wavelen gth is* $2\,440\,\mathrm{cm^2{\cdot}mmol^{-1}}$. *You should give your answer in millilitres.*

Kiki will once become a real pharmacis

Problem FoL.31 ... valuing the decays

An isotope of gold ^{173}Au *has a half-life of* $T_{Au} = 59.0\,\mathrm{ms}$. *It decays into an iridium isotope* ^{169}I
by emitting an alpha particle. This iridium isotope has a half-life of $T_{Ir} = 0.400\,\mathrm{s}$ *and decays int*
^{165}Re. *Initially we had a pure sample of gold isotope* ^{173}Au. *Find the time at which the amoun*
of gold in our sample will be the same as the amount of iridium. You can assume that masse
of isotopes are directly proportional to their nucleon numbers. *Karel was thinking hard*

Problem FoL.32 ... Flash

After an attack, the superhero Flash does not bother to stop. Instead, he circles around th Earth, and attacks again. Initially, his speed was $v = 0.8c$. During the first attack, he looses ha of his momentum. After repeating the run around the Earth with his new speed and engagin in the second attack, he looses half of his momentum again. What is the ratio E_1/E_2 of energie released during the first and the second attack? According to dc.wikia.com, Flash's rest mas is $m_0 = 89\,\mathrm{kg}$.

Mirek watching superhero programmes

Problem FoL.33 ... we need to go deeper

What is the moment of inertia (relative to its axis of symmetry o) of the lamina shown in th
igure (the lamina contains the shaded parts only)? The shape is constructed in the followin
way: given a semicircle, we cut out semicircular holes of it with their radii being half of th
adius of the original semicircle. We then insert four four-times smaller semicircles (two int
ach hole) in these holes, again with semicircular holes cut out of them and in these we the
nsert smaller semicircles, continuing ad infinitum. The mass of the plate is $m = 7\,\mathrm{kg}$*, and th*
adius of the largest semicircle is $R = 40\,\mathrm{cm}$*.*

Xellos was thinking about the old competition days

Problem FoL.34 ... little skewer

A thin rod of length $2l = 30\,\mathrm{cm}$ is placed into a hemispherical bowl of radius $R = 10\,\mathrm{cm}$ Assuming that the rod is at equilibrium, what is the angle α (in degrees) between the rod and the vertical?

f(Aleš) really liked a rod problem on the internet

Problem FoL.35 ... tesseract

An expedition exploring the ocean floor has found a four-dimensional cube called tesseract o hypercube. Naturally, the explorers decided to investigate its physical properties, so they trie to melt it. They found out that it was made of an isotropic material with a large coefficien of linear thermal expansion which, in addition, was found to grow linearly with increasin temperature. Specifically, $\alpha_a(T_1) = 5 \cdot 10^{-4}\,\mathrm{K}^{-1}$ and $\alpha_a(T_2) = 2 \cdot 10^{-3}\,\mathrm{K}^{-1}$ for $T_1 = 300\,\mathrm{K}$ and $T_2 = 400\,\mathrm{K}$ respectively. Find the percentage increase in the 4-volume of the hypercub when heated from T_1 to T_2 given that at the temperature T_1, its edge length is $a = 10\,\mathrm{cm}$

Mirek was thinking about the physics in Avengers

Problem FoL.36 ... short hands

Mirek wanted to measure the dimensions of his room but all he could use was a 2 metres lon
olding rule. With each of his hands taking hold of one end of the ruler, he found out that t
is surprise he could not straighten the ruler by spreading out his arms. What is the vertica
listance between the lowest point of the sagged ruler and the horizontal line joining Mirek'
ands? The ruler consists of four identical sections of length $l = 0.5\,\mathrm{m}$ *and Mirek's arm spa*
s $L = 1.8\,\mathrm{m}$*. You should neglect friction and overlaps between the individual sections of th*
uler. *Mirek discovered that height and the arm span are more or less the same*

Problem FoL.37 ... resistant Fibonacci

Let us consider a configuration of resistors with identical resistances $R = 1\,\Omega$ built as follows: first we connect two resistors in series, next to them we add two resistors connected in parallel, followed by 3 resistors in parallel, then 5, 8, 13, ... resistors in parallel until we get a Fibonacci sequence of resistors. Find the resistance of such a configuration. *Mirek could not resist*

Problem M.1 ... stilettos

Which of the following bodies exerts a greater pressure on the ground and what is its value
A cube made of steel, its side $3\,\mathrm{m}$ *long or a woman weighing* $59.\bar{9}\,\mathrm{kg}$ *wearing high heels wit*
a diameter of $5\,\mathrm{mm}$*? Consider the situation when the entire weight of the woman rests on on*
heel. *Monika stepped on a bug*

Problem M.2 ... traffic jam

Consider a car moving slowly in a traffic jam with a velocity of $v = 2\,\mathrm{m{\cdot}s^{-1}}$*. Find the rate c*
change of an angle under which a road sign placed $2.5\,\mathrm{m}$ *above the ground is seen from the ca*
at an initial distance of $x = 30\,\mathrm{m}$ *from the sign. The dimensions of the car and the road sign ar*
not to be taken into account. *Verča was observing the behaviour of drivers in a traffic jam*

Problem M.3 ... very slow relativity

Consider two balls moving with velocities $0.4\,\mathrm{m{\cdot}s^{-1}}$ *and* $0.9\,\mathrm{m{\cdot}s^{-1}}$ *in the same direction. Befor the collision, the slower ball moves ahead of the faster one and after the collision the balls stic together. Find out by how much the temperature of the balls will increase immediately afte the collision, assuming that their temperatures before the collision were equal. The specifi heat capacity of the material of the balls is* $c = 0.02\,\mathrm{mJ{\cdot}K^{-1}{\cdot}g^{-1}}$*. The balls are known to hav identical masses.*

Jančі was trying to symplify Lukáš's problem on relativity

Problem M.4 ... splash

Find the minimal height above the water level at which we have to place the bottom end of a vertically oriented rod with length $l = 50\,\mathrm{cm}$ in order for the entire rod to be submerged under the water when it is dropped. The density of the rod is half the density of water and the rod is slightly weighted at its bottom so that we don't have problems with stability.

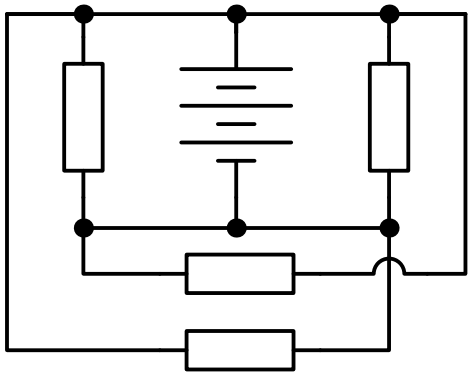
Lukáš was taking a bath

Problem E.1 ... flash

The battery in Janči's camera has the following specifications written on it: 3.6 V *and* 1 250 mAh
The capacitor used in the flash has a capacity of 90 μF *and it is charged to the voltage of* 180 V
each time the flash is used. Assume that during the process of charging the flash, exactly ha
of the energy is lost. How many pictures with flash can the camera shoot before the batter
dies (assuming that initially, the battery was fully charged)? You can assume that the batter
s ideal in the sense that its voltage does not decrease during its usage. A fair warning: th
esult should be an integer. *Janči hates taking pictures with flash*

Problem E.2 ... tangled resistances

Find the current I through the source with voltage $U = 1\,\mathrm{V}$ assuming that each resistor ha
esistance $1\,\Omega$. *Jančі was trying to think of a task, but not about the solution*



Problem E.3 ... aberration

Light with a wavelength of 400 nm *in vacuum has a wavelength of* 265 nm *when it passes throug*
a lens, while light with a wavelength of 700 nm *in vacuum has a wavelength of* 460 nm *when i*
passes through the same lens. Using blue light (vacuum wavelength 400 nm*), the focal length o*
the lens is 1 m*. By how much will the focal length change if we use red light (vacuum wavelengt*
700 nm*) instead? Assume that the lens is thin.*

Jančі trying to remember the parts of optics he actually liked

Problem E.4 ... useless work

Consider a parallel-plate capacitor charged to a voltage of $1\,\mathrm{V}$*. In a direction perpendicular t*
he electric field lines between the plates, there is a magnetic field of $2\,\mu\mathrm{T}$*. Distance between th*
wo plates is $d = 0.1\,\mathrm{mm}$*. Assume that an electron is emitted from the plate at lower potentia*
with zero initial speed. Find the speed at which it arrives at the second plate.

Xellos likes to be in the way

Problem X.1 ... let's spread out

At some temperature T, the cubic lattice of an iron crystal undergoes a transition from body-centred to face-centred phase. During this transition, the length of the unit cell edge increases by 22 %. Find the factor by which the density of the crystal will decrease.

Tomáš Bárta was thinking about iron

Problem X.2 ... heating season

How many times do we have to let a bouncy ball with a mass of $m = 200\,\mathrm{g}$ *fall on the groun made of polyvinyl chloride with a density of* $\varrho = 1380\,\mathrm{kg{\cdot}m^{-3}}$ *and a specific heat capacit* $= 0.9\,\mathrm{kJ{\cdot}kg^{-1}{\cdot}K^{-1}}$*) from the height* $h = 1\,\mathrm{m}$ *so that a thermally well-isolated piece of th ground with area* $S = 1\,\mathrm{m^2}$ *and thickness* $d = 1\,\mathrm{cm}$ *would increase its temperature by on degree of Celsius? The coefficient of restitution* k *for collisions between the bouncy ball an the ground, defined as the ratio of the speed of the bouncy ball after a collision and its spee before the collision, is known to be* $70\,\%$*. Use the following value for the acceleration due t gravity:* $g = 9.81\,\mathrm{m{\cdot}s^{-2}}$*.*

Terka trying to warm her frozen feet by alternative means

Problem X.3 ... hexagonal sphere

The lattice of an alpha-boron nitride crystal consists of atomic layers in which the atoms ar
rranged in a hexagonal structure. We managed to acquire a sphere with radius $r = 1\,\mu\text{m}$ *mad*
of this material. Find the maximum number of layers which can be intersected by a straigh
ine intersecting the sphere. You are encouraged to look up any information needed.

Xellos having nightmares about chemistry

Problem X.4 ... firm or springy

Pistons in the fork of a bicycle have a cross-section of $S = 5\,\mathrm{cm}^2$ *and height* $h = 20\,\mathrm{cm}$*. Assum hat a manometer connected to the fork gives a reading of* $p_1 = 100\,\mathrm{psi}$*. By how much will th ork be compressed (in* cm*), when a* $m = 60\,\mathrm{kg}$ *man leans on it with all his weight? Assum deal diatomic gas behaviour and that an adiabatic process takes place. Do not forget that th ike fork has two arms.* *Mirek's been avoiding fixing his bike for some time already*